

Big Data Foundation

Syllabus

Lead Author: Mario Bojilov



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Scope and Purpose of this Document

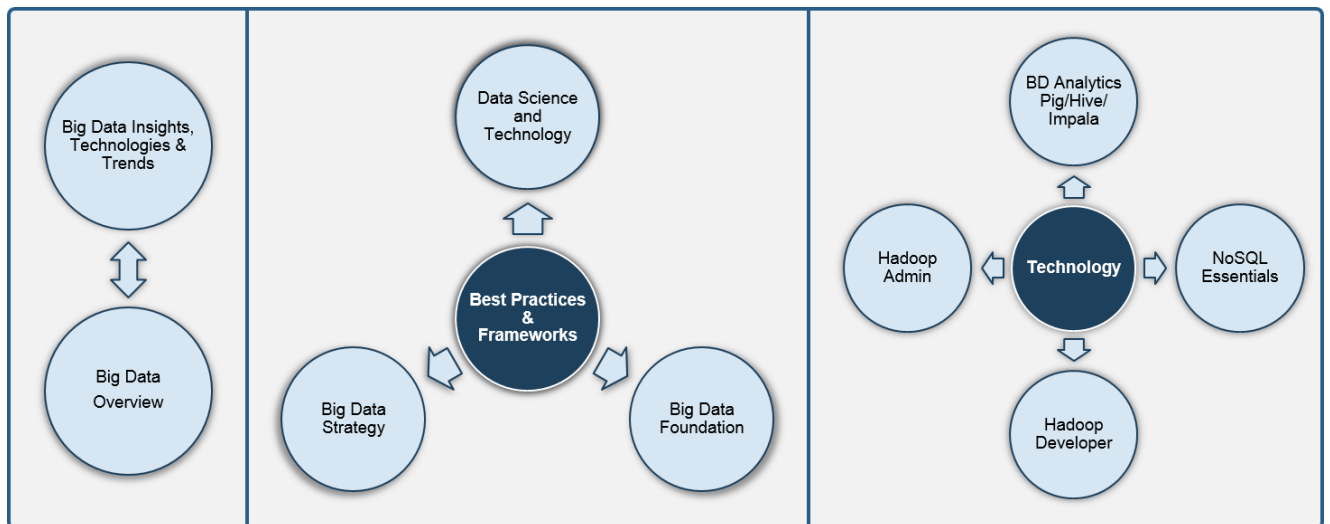
Synopsis

The purpose of this document is to inform all parties interested in the Big Data Foundation course of the areas covered in the Big Data Foundation certification.

The target audience for this document is:

- (Potential) Examination candidates
- (Aspiring) Instructors
- Accredited Training Organizations

Big Data Qualification Scheme



Foundation Qualification

Level of Difficulty

Bloom Level 1 & 2: Knowledge and Comprehension

Qualification Objectives

- Explain Big Data, its origin, and its characteristics.
- Discuss about the tools applicable to Big Data processing.
- Explain data mining.
- Discuss the most popular Big Data technologies—Hadoop and MongoDB.
- Tell about Big Data projects and the main players involved.

Format of the Examination

- Exam Format: Closed-book format. Web-based
- Questions: 40 multiple-choice questions
- Passing Score: 65%
- Exam Duration: 60 minutes. An additional 15 minutes is available to non-native English speakers
- Proctoring: Live / Webcam

Syllabus Areas – Big Data Foundation

1. Course Introduction

- 1.1 Course Learning Objectives
- 1.2 Course Agenda
- 1.3 Activities
- 1.4 Exam
- 1.5 Course Book
- 1.6 Cloud Credential Council® (CCC)
- 1.7 Certification Value

2. Big Data Fundamentals

2.1 Big Data – History, Overview, and Characteristics

- History
- Big Data Overview – Definition
- Big Data Overview – Benefits
- Big Data Characteristics – Volume
- Big Data Characteristics – Velocity
- Big Data Characteristics – Variety

2.2 Big Data Technologies – Overview

- Hadoop – Introduction
- Hadoop – Usage Example
- Hadoop – Concepts
- Hadoop – Use Cases
- MongoDB – Introduction
- MongoDB – Features
- MongoDB – Concepts
- MongoDB – Use Cases

2.3 Big Data Success Stories

- Target US – Background
- Target US – Analysis
- Fighting Piracy on High Seas – Background
- Fighting Piracy on High Seas – Analysis

2.4 Big Data – Privacy and Ethics

- Privacy – Compliance
- Privacy – Challenges
- Privacy – Approach
- Ethics

2.5 Big Data Projects

- Who Should Be Involved?
- What Is Involved?

3. Big Data Sources

3.1 Enterprise Data Sources

- Enterprise Systems
- Oracle
- SAP
- Microsoft
- Data Warehouses
- Unstructured Data – Introduction
- Unstructured Data – Metadata

3.2 Social Media Data Sources

- Introduction
- Facebook – Introduction
- Facebook – Public Feed API
- Facebook – Keyword Insights API
- Facebook – Graph API
- Twitter – Introduction
- Twitter – Streaming APIs
- Twitter – REST APIs
- Other Social Media

3.3 Public Data Sources

- Introduction
- Weather
- Economics
- Finance
- Regulatory Bodies

4. Data Mining – Concepts and Tools

4.1 Data Mining – Introduction

- Introduction
- Types of Data Mining – Overview
- Types of Data Mining – Classification
- Types of Data Mining – Association
- Types of Data Mining – Clustering

4.2 Data Mining – Tools

- Introduction
- Weka
- Modules of Weka Applications
- KNIME
- KNIME – Example
- R Language

5. Big Data Technologies – Hadoop

5.1 Hadoop Fundamentals

- Introduction
- Main Components of Hadoop
- Additional Components of Hadoop

5.2 Install and Configure

- Download
- How to Install and Configure

5.3 MapReduce

- Introduction
- How Does It Work?

5.4 Data Processing with Hadoop

- Introduction
- Twitter Sentiment Analysis – Overview
- Twitter Sentiment Analysis – Algorithm
- Network Log Analysis – Overview
- Network Log Analysis – Algorithm

6. Big Data Technologies – MongoDB

6.1 MongoDB Fundamentals

- Introduction
- Replication
- Sharding
- Sharding and Replication
- MongoDB Ecosystem – Languages and Drivers
- MongoDB Ecosystem – Hadoop Integration
- MongoDB Ecosystem – Tools

6.2 Install and Configure

- Download
- How to Install and Configure

6.3 Document Databases

- Introduction
- Documents
- Document Design Considerations
- Fields

6.4 Data Modelling with Document Databases

- Introduction
- Twitter Sentiment Analysis
- Twitter Sentiment Analysis – Algorithm
- Network Log Analysis
- Network Log Analysis – Algorithm

References

Specific Sources:

- Facebook.com
- Twitter.com
- MongoDB.org
- Hadoop.org
- New York Times – “How Companies Learn Your Secrets” (Target's Story – Module 1)
- Business Insider (Pirates Story – Module 1)
- SAP.com
- Microsoft.com
- Oracle.com
- International Monetary Fund – <http://www.imf.org/external/data.htm#data>
- World Bank – <http://data.worldbank.org/>
- US Government – <http://www.nber.org/data/>
- European Commission – http://ec.europa.eu/economy_finance/db_indicators/
- Open Data US – data.gov
- Open Data Australia – data.gov.au
- Open Data European Union – open-data.europa.eu/en/data/
- KNIME – [knime.org](http://www.knime.org)
- Weka – <http://www.cs.waikato.ac.nz/ml/weka/>